SCREAMINGFACE · URL4 · DESIGN DECISION

url4 topology — node, host, discovery, addressing, transport

Sharp definitions before the Engine grows further

STATUS	proposed — owner review; one decision left open in §5
WORK ITEM	OME-1110
DATE	2026-09-04
OWNER	Sergey Bershadsky (execution and telemetry architecture)
GRAMMAR OWNER	Kevin McDonough (URL4 spec, Parts A/B)
SPEC POINTERS	URL4.ai Specification v0.5 DRAFT, Parts A (§1) and B (§2–§8); pre-v0.5 monolith v0.2 for §9–§40

Read §0 first. Everything else is the reasoning behind one line of that table.
One decision is left open for the owner, in §5.

0. The eight answers

This document fixes the words we use for the pieces of url4. It is short on purpose. Each section states a definition, says where the spec agrees or is silent, and stops.

#	QUESTION	ANSWER
1	What is a node?	A stateless function behind a path. It takes `(context)!intent` and returns a result. Every node can evaluate a full url4 expression.
2	What is a host?	The origin (`scheme://authority`) that mounts one or more nodes at paths and answers discovery. `/` is its default node.
3	Swarm? Composition?	Not protocol words. The expression **is** the composition. The hosts it touches are its request tree. “Swarm” is only a deployment word for one operator’s hosts.
4	`.well-known` or OPTIONS?	Both are documented in §5 with their trade-offs. The owner picks there.
5	`/name` vs `url4://name` ?	`/name` is a node mounted on the host that is evaluating. `url4://name` is host `name` ’s default node `/` . Spec Part B §5.4 already says this.
6	Where does an ensemble run?	Wherever an evaluator runs: the SDK on a laptop, or a host. A local host may mount remote nodes under local names (proxy mounts).
7	Can I test a host first?	One fetch of the host’s capabilities document, or one OPTIONS per node (§5). A separate “plan” concept is deferred (§9).
8	Streaming fallback?	One GET asks for the richest mode; the node answers with the best it has: WebSocket, then SSE, then sync (§6). Sync is the only MUST. Async is a separate axis.

What changes against today

- The Engine becomes a url4 **host**. Today no url4 node is reachable over HTTP; the SDK’s node-to-node code is unused.
- Every node speaks the same GET. The node picks the richest delivery it supports: **WebSocket → SSE → sync**, in one round trip. Sync is the only MUST.
- Two spec words move: the spec’s Node becomes our Host; the spec’s Endpoint becomes our Node. Appendix A asks Kevin for the rename.

1. Terms

Eight words. Each has one meaning.

OUR WORD	MEANING	SPEC WORD TODAY	IN THE SDK	IN THE ENGINE
Node	A stateless function at a path. Accepts `GET <path>?q=<expr>` , evaluates it, returns a result plus envelope.	Endpoint (Part A §1.4)	an entry in `Url4Node` ’s endpoint registry	a route such as `/anthropic/<model>`
Host	An origin that mounts nodes at paths, serves `/` as the default node, and answers discovery.	Node (Part A §1.4)	`Url4Node` (the class name lags; Appendix B)	the App, after this change
Mount	The binding of a path to a node implementation: `local` (in-process), `command` (subprocess), or `proxy` (forwards to a declared remote node).	not defined	`endpoint()` , `[commands]` , —	in-process handlers only

OUR WORD	MEANING	SPEC WORD TODAY	IN THE SDK	IN THE ENGINE
Evaluator	Whatever runs an expression: resolves sources, fans out, reduces, runs the intent. Every node contains one.	“the node executes” (Part A §1)	`url4.dag.run`	`Url4Executor` in the Runner
Intent processor	The thing inside a node that turns resolved context plus intent into a result: a model call, a script, a command.	Intent processor (Part A §1.4)	endpoint handler	one `httpx` call to aigateway
Requestor	Whoever sends an expression.	Requestor (Part A §1.4)	`Client`	product client
Request tree	The hosts and nodes one expression touches. Strict tree, never a graph (v0.2 §16.1).	“request tree”, “call tree” (v0.2 §22)	the compiled DAG, per node	one run
Capabilities document	JSON a host publishes to say which nodes, mounts, features and delivery modes it offers.	named in Part A §1.4; schema unwritten (Part G)	none	none

Words we do not use in the protocol: ensembler, orchestrator, swarm, composition, plan, fusion. “Fusion” stays product copy.

2. Addressing

Three address forms. The spec fixes their meaning in Part B §3.1.1 and §5.4; we add one rule about the root.

- `/claude(ctx)!'go'` — a node on the host that is evaluating. Resolves to `url4://<current host>/claude` (Part B §5.4).
- `url4://beta.example/claude(ctx)!'go'` — a node on another host. That host evaluates the sub-expression itself (Part B §3.1.1; SDK invariant OME-535).
- `url4://beta.example` — the host’s default node. **Our rule:** `url4://host` ≡ `url4://host/`.
- `https://data.example/rows` — plain HTTP data. No url4 headers, no envelope, `@` is an error (Part B §3.5).

Root and version. `/` is the host's default node. The spec makes `/v1` the protocol-version path (v0.2 §35.1). We keep both: `/` serves the current version; `/v1` is an explicit alias. The SDK's `url4 serve` (`/v1`) and `Client` (default `/v1`) move to `/` (Appendix B).

Scheme inheritance. A bare relative data URI inherits the scheme of the request that carried it (Part B §5.4.1). Under `url4://` it is a url4 read; under `https://` it is a plain read.

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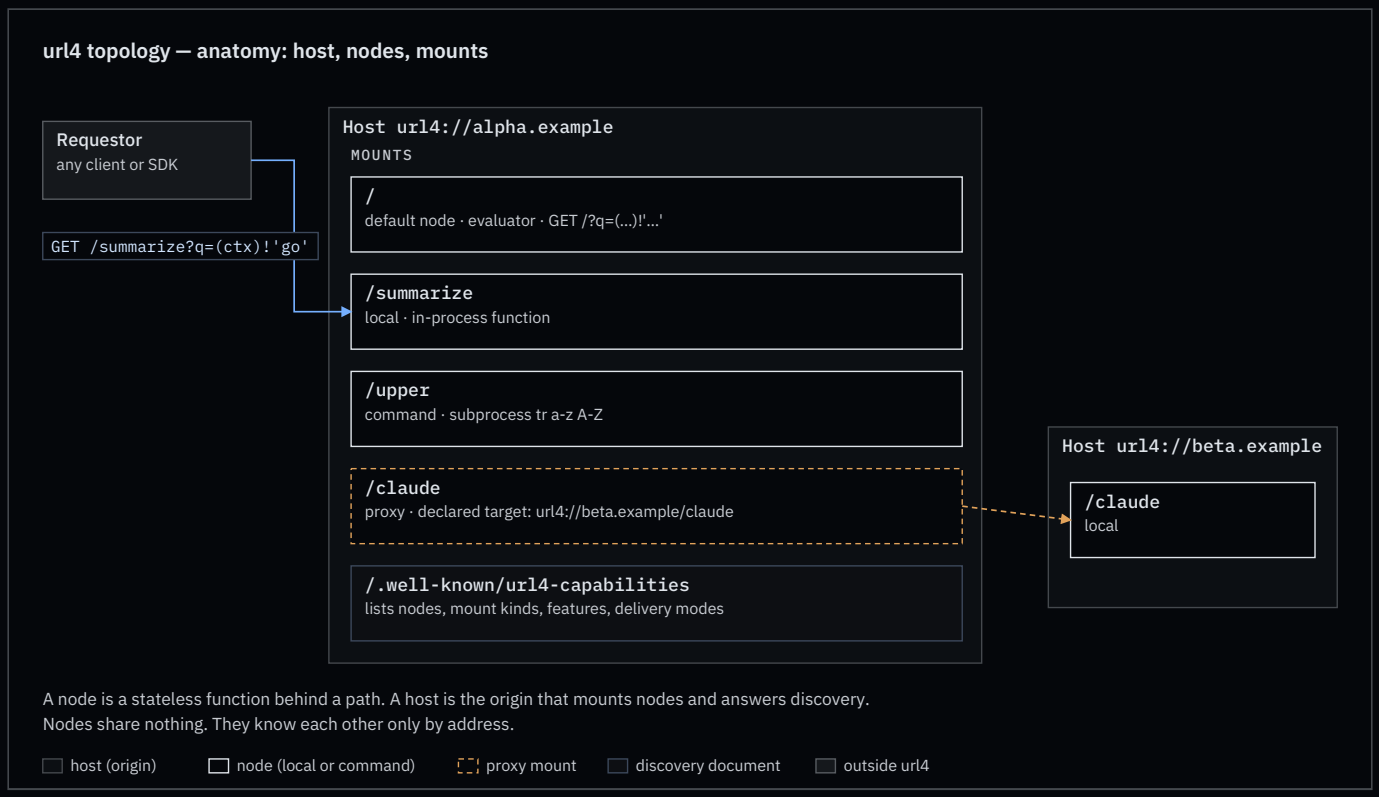
3. Node contract

A node is a function. That is the whole idea.



A node:

- **MUST** answer `GET <path>?q=<expression>` and evaluate the expression. Every node evaluates url4; a node may walk a DAG before it reaches its own intent processor. There is one grade of node, not two.
- **MUST** be stateless per invocation. It keeps nothing between calls. Its own data is reached through `@` (Part B §5.6) and lives behind it, not in it. Multi-turn work uses an explicit session key (`;coord=`, Part B §4.2), never process memory.
- **MUST** return the envelope (v0.2 §13) and state the delivery mode it actually used (v0.2 §11.4).
- **MUST NOT** resolve `@` inside a sub-expression addressed to another host (Part B §5.6.3.1).
- **MUST** answer sync. **SHOULD** offer stream (SSE), WebSocket, and async, and advertise which (§5, §6). When asked for more than it has, it answers with the richest mode it does have; that is never an error.
- **MAY** be mounted on any host, or be its own origin. A Lambda-style function with its own URL is a host with one node.
- Speaks GET. A WebSocket, when offered, is the same GET with `Upgrade: websocket` (§6). Other verbs are 405, except the ones this document adds: `DELETE` on a run handle (§6) and, if adopted, `OPTIONS` (§5).

Nodes share nothing. A node knows another node only by address, and only because the expression named it.

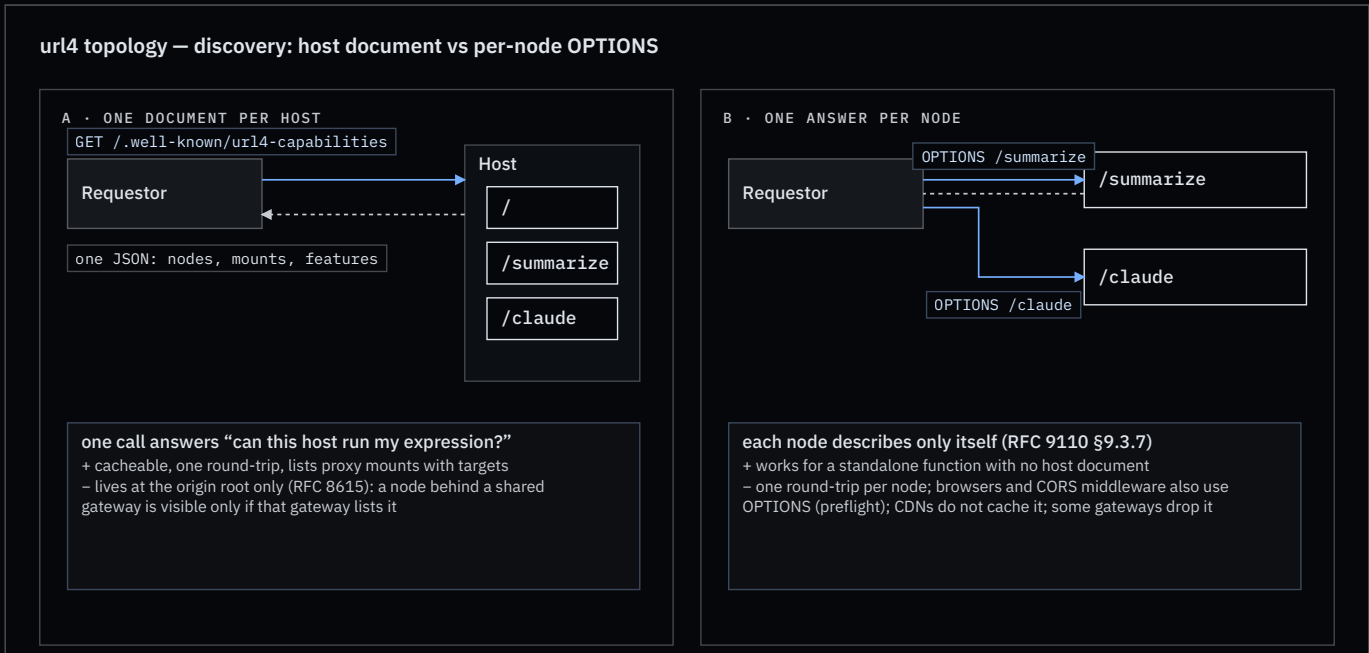
4. Host contract

A host is an origin. It routes; it does not evaluate.

- Mounts one or more nodes at paths. `/` is the default node.
- Mount kinds: `local` (a function in the process), `command` (a subprocess; doctrine N4), `proxy` (forwards the sub-request verbatim to a declared target on another host).
- Declares proxy mounts with their targets in its capabilities document. Attribution and consumer disclosure (v0.2 §16.2.2) need the real target, so proxies are never hidden.
- Serves discovery (§5) and the version alias `/v1` (§2).
- A local SDK process that mounts remote nodes under local names is a host. That is how “run the ensemble on my laptop, but call `/claude` as if it were mine” works.

5. Discovery — the open decision

Two ways for a requestor to learn what a host can do. Neither exists yet: the spec names the capabilities document (Part A §1.4) but its schema is in the unwritten Part G; OPTIONS appears nowhere in the spec.



They can coexist: the host document indexes, OPTIONS answers for one node. Section 5 leaves the choice to the owner.
Neither exists yet: the spec names the document (Part A §1.4) but its schema is in the unwritten Part G; OPTIONS appears nowhere.

	A · HOST DOCUMENT `GET /.well-known/url4-capabilities`	B · PER NODE `OPTIONS /<node>`
“Can this host run my expression?”	one call, whole answer	one call per node named in the expression
Standalone function with its own origin	works: the document sits at that origin’s root	works
Node behind a shared, non-url4 gateway	invisible unless the gateway lists it (RFC 8615: root of the origin only)	works: the node answers for itself
Caching	ordinary GET: `ETag`, CDN, browser cache	not cached by CDNs or browsers

	A · HOST DOCUMENT `GET /.well-known/url4-capabilities`	B · PER NODE `OPTIONS /<node>`
Browsers and CORS	plain GET	preflight uses the same verb; CORS middleware often answers first (RFC 9110 §9.3.7 allows a body, browsers ignore it)
Gateways and serverless platforms	fine	some strip or auto-answer OPTIONS
Proxy mount targets (attribution)	listed in one place	per node
Spec status	named; schema to write (Part G §27.2)	not in the spec

Both share one hazard: the Engine's JWT header is called `URL4-Capability`. The document is "capabilities". Appendix B proposes renaming the header.

Draft capabilities document (host level; a node's OPTIONS answer is the one entry for that node):

```
{
  "url4": "0.5",
  "host": "url4://alpha.example",
  "default_node": "/",
  "nodes": {
    "/":          { "mount": "local",   "delivery": ["sync", "stream"],
                    "features": ["nesting", "iteration", "broadcast", "self_ref"] },
    "/summarize": { "mount": "local",   "delivery": ["sync"] },
    "/upper":     { "mount": "command", "delivery": ["sync"] },
    "/claude":    { "mount": "proxy",   "target": "url4://beta.example/claude" }
  },
  "holdings": { "collections": ["default", "science"], "self_ref_support": true }
}
```

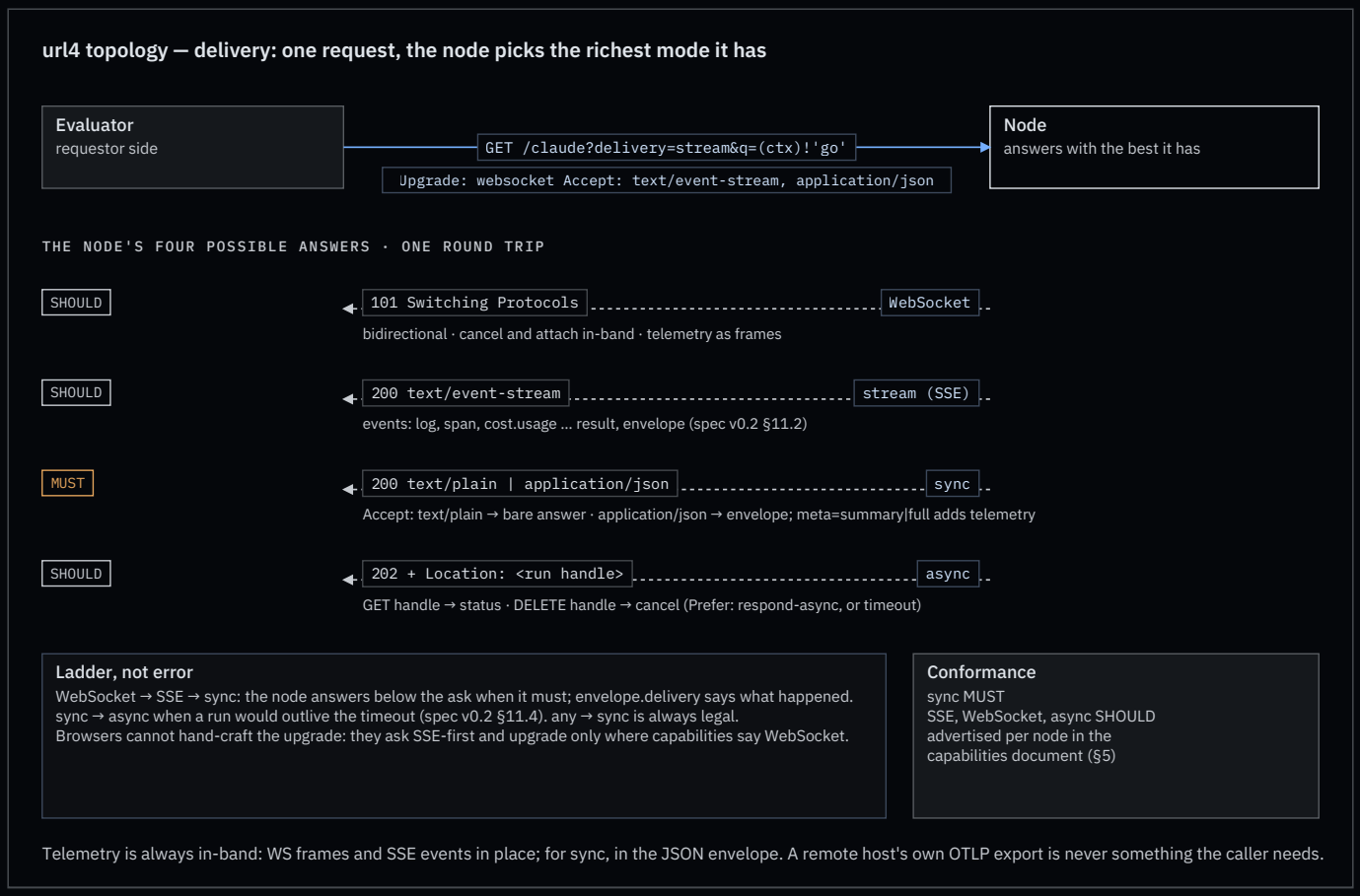
Owner decision (open). Pick one:

- ☐ **A only** — host document is a MUST; nothing per node.
- ☐ **B only** — OPTIONS per node is a MUST; no host document.
- ☐ **A MUST + B SHOULD** — the document indexes the host; a node may also answer OPTIONS with its own entry.

Recommendation: A MUST + B SHOULD. A alone fails the standalone-function-behind-a-gateway case; B alone fails the single-call test in answer 7.

6. Delivery and transport

The spec defines three delivery modes and a fallback rule (v0.2 §11). We adopt them, add WebSocket as a fourth answer, and make the whole ladder one request.



One request, the node picks. The evaluator sends the richest ask it can, once:

```
GET /claude?delivery=stream&q=(ctx)!'go'
Upgrade: websocket
Accept: text/event-stream, application/json
```

The node answers with the best it supports. RFC 6455 lets a WebSocket handshake ride an ordinary GET; RFC 9110 lets a server ignore `Upgrade` and answer normally. So one round trip covers the whole ladder:

NODE ANSWERS	MEANING	TELEMETRY TRAVELS AS
`101 Switching Protocols`	**WebSocket**: bidirectional; cancel and attach in-band	frames, same names as the SSE events
`200 text/event-stream`	**stream**: SSE on the same GET (v0.2 §11.2)	SSE events, then `result`, then `envelope`
`200 text/plain` or `application/json`	**sync**: bare answer, or the envelope, by `Accept`	envelope `meta` (below)
`202 + Location`	**async**: run handle; `GET` it for status, `DELETE` it to cancel	on the handle

Rules:

- **sync is the only MUST.** SSE, WebSocket and async are SHOULD, advertised per node in the capabilities document (§5). A serverless function that offers only sync and SSE is conformant.

- **Ladder, not error.** WebSocket → SSE → sync. A node answering below the ask is normal; the envelope's `delivery` field says what happened (v0.2 §11.4).
- **Async is an axis, not a rung.** It is requested with `Prefer: respond-async`, or entered by the node when a sync run would outlive the timeout (v0.2 §11.4 `sync → async`).
- **Browsers go SSE-first.** A browser cannot attach `Accept` to a WebSocket handshake, so a browser evaluator asks for SSE and upgrades only where the capabilities document says WebSocket is offered.
- **The Engine's product session** keeps its WebSocket at `/ws`. It is one instance of the WebSocket rung, not a separate protocol.
- **Telemetry is always in-band.** Whatever the mode, the caller receives logs, spans and cost on the same connection or in the envelope. A remote host's own OTLP export is its business, never something the caller depends on (§7).

Telemetry per mode. WebSocket and SSE carry it in place: an SSE body is a sequence of named events (`event: log`, `event: span`, `event: cost.usage`, then `event: result`, `event: envelope`), in order, on the one response. The spec already builds on this (v0.2 §12.5 is an SSE event catalog); our three signals are three more names. What SSE lacks against WebSocket is only the return path: no in-band cancel or attach.

Sync has no room for that: the body is the answer. So the wrapper is negotiated, with two knobs that must not be conflated:

KNOB	GOVERNS	VALUES
`Accept` header	the wrapper	`text/plain` (default): the bare answer, no logs, no telemetry, what `url4 serve` returns today. `application/json`: the envelope (v0.2 §13)
`meta` param	how much the envelope carries	`none`: result and status. `summary`: counts, latency, cost. `full`: per-source detail, nested child envelopes (v0.2 §13.3), and a `telemetry` block with logs and spans
`fmt` param	the shape of the answer content	`text`, `markdown`, `json` (v0.2 §7.1); orthogonal, a JSON envelope can wrap a markdown answer

A JSON answer with no `meta` gets `summary`, so asking for JSON always buys the cheap insight; `meta=full` is the explicit debug switch. What a node exposes is its decision (v0.2 §14.4): a production node may answer `meta=full` with the `telemetry` block redacted, but it MUST keep the structure and use `null` or `"redacted"`, never drop fields silently. A development node hands over everything. Same envelope, same evaluator code, different policy.

Cancel. The spec calls cancellation a gap (v0.2 §34). Over WebSocket, cancel is in-band. Otherwise it is `DELETE` on the run handle returned as `Location` (and `Link rel=self`), which the Engine already does. New terminal state `cancelled`; SSE event `request.cancelled`.

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7. Telemetry

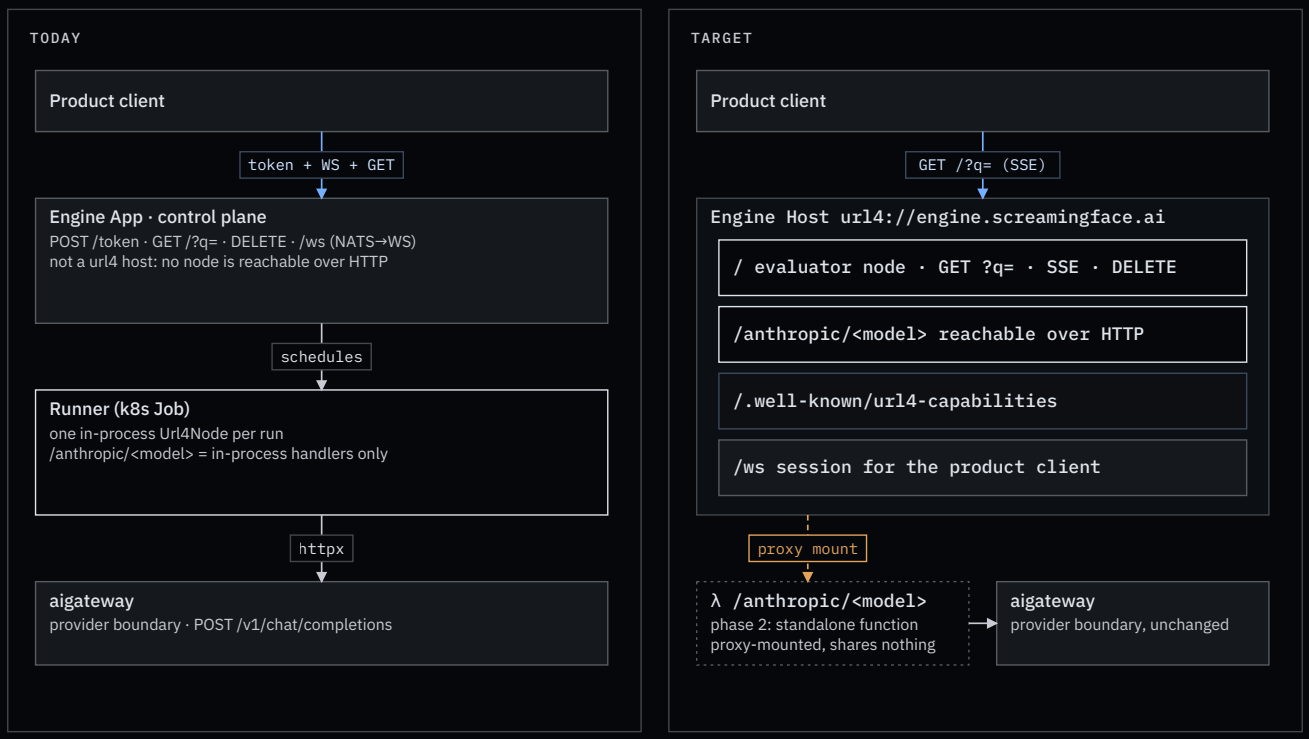
Three signals, one trace, as in the doctrine skill. What changes is where a one-shot GET puts them.

- **sync**: in the body, only when the caller asked for the envelope (`Accept: application/json`, §6). `meta` sets the level (`summary` by default, `full` adds logs and spans; v0.2 §13.2). At `meta=full` a child's envelope nests in `source.envelope` (v0.2 §13.3). Each node aggregates only what it saw itself (v0.2 §14.1). A bare `text/plain` answer carries nothing.
- **stream** and **WebSocket**: as events: logs, spans, `cost.usage` (self and subtree), then `result`, then `envelope`. Same names in both.
- **durable**: OTLP export to the trace backend. There is no separate "Enclave" store; the exporter superseded it.

This closes doctrine item F4. **Idempotency:** a url4 GET is idempotent in the HTTP sense (RFC 9110 §9.2.2): repeating it has no extra server-side effect. Results need not be identical; the spec's §33 conflates the two (Appendix A).

8. The Engine as a host

url4 topology — the Engine today and as a url4 host



The Runner becomes the evaluator process the / node spawns per run. The App keeps /ws for the product client and gains the node surface.
Serverless posture: nodes are functions; the host is the only thing that needs an origin.

	TODAY	TARGET
App	control plane: `POST /token`, `GET /?q=`, `DELETE`, `/ws` bridged from NATS	a **host**: `/` evaluator node (`GET ?q=` with WebSocket, SSE, sync and async answers, `DELETE`), model nodes over HTTP, capabilities, `/ws` kept for the product client
Runner	k8s Job with one in-process `Url4Node`; model routes are in-process handlers	the evaluator process the `/` node spawns per run
Model nodes	`/anthropic/<model>` reachable only inside the Runner	reachable over HTTP on the host; later each one a standalone function, proxy-mounted
aigateway	provider boundary	unchanged
Node-to-node	unused SDK code	the normal path

Phases. (1) Host surface on the App: discovery per §5, `GET /?q=` answering SSE and sync (WebSocket via `Upgrade` where the App already has it), `DELETE`. (2) Model nodes over HTTP. (3) Standalone model functions, proxy-mounted, listed in the capabilities document.

Serverless posture. A node is a function; it needs no origin of its own. Only the host needs one. The Runner is already function-shaped. The App is not: it holds an audience count in memory, runs two perpetual tasks, and gates runs on an attached WebSocket. Those belong to the product session, not to the node surface, and stay in the App.

9. Deferred

- **Plan / preflight** as a defined concept. Today the SDK's `Graph.validate()` checks syntax only and the Engine's preflight checks routes on one host. Deferred by owner decision; §5 answers the practical question.

- **Swarm** as a protocol noun. Would need membership and trust rules that belong to the unwritten governance parts.
- **Node grades** (processor-only vs evaluator). Rejected: every node evaluates.

Appendix A — proposed spec deltas for Kevin

Each item names the anchor and the change. All are proposals.

1. **Part A §1.4 — rename.** Node → **Host** ("an origin implementing the protocol; mounts nodes"). Endpoint → **Node** ("a path on a host bound to an evaluator and an intent processor"). Add **Mount** (`local | command | proxy` , proxies declare their target).
2. **Part B §3.1.1, v0.2 §35.1 — root.** `url4://host` ≡ `url4://host/` ; `/` is the host's default node at the current version; `/v1` is a version alias.
3. **Part C §11.2 — bindings.** `delivery=stream` is SSE on the same GET, requested with `Accept: text/event-stream` . A node MAY also offer WebSocket by honouring `Upgrade: websocket` on that same GET (`101`); frames carry the same event names.
4. **Part C §11.4 — ladder and floor.** The node answers with the richest mode it supports, WebSocket → SSE → sync, in one round trip; sync is the only MUST and `any → sync` is always legal. A node answering below the ask is not an error; the envelope's `delivery` field says what happened.
5. **v0.2 §33 (Part H in v0.5) — idempotency.** Replace "the protocol does not guarantee idempotency" with: GET is idempotent per RFC 9110 §9.2.2; results are not guaranteed deterministic.
6. **v0.2 §34 (Part H in v0.5) — cancellation.** `DELETE` on the run handle (`Location`). Terminal state `cancelled` ; SSE event `request.cancelled` ; propagates to in-flight children.
7. **Part D §13 — envelope for sync callers.** `Accept: text/plain` returns the bare answer; `Accept: application/json` returns the envelope, `meta=summary` by default. At `meta=full` the envelope carries a `telemetry` block (`logs[]` , `spans[]` , `cost`) that a node MAY redact per §14.4 (structure kept, `"redacted"` sentinel). `fmt` stays the answer's content format and is orthogonal.
8. **Part G §27.2 — capabilities document.** Schema as drafted in §5: `nodes` keyed by path, `mount` , `target` , `delivery` , `features` ; `holdings` . Optional binding: the same entry as an `OPTIONS` body with `Content-Type: application/url4-capabilities+json` .

Appendix B — follow-up work items

To file after owner review, one per landing.

LANDING	ITEM
screamingface-engine	Host surface phase 1: discovery per §5, `GET /?q=` with SSE, `DELETE` on the run handle
screamingface-engine	Model nodes reachable over HTTP (phase 2); standalone model functions, proxy-mounted (phase 3)
screamingface-engine	Rename the `URL4-Capability` JWT header to avoid the "capabilities" collision
url4-sdk	Delivery negotiation in `Url4Node.asgi()` : `Upgrade` / `Accept` handling, SSE body, optional WebSocket answer; evaluator-side single-request ladder in `HttpIOLayer`
url4-sdk	Capabilities document and/or OPTIONS responder, after the §5 decision; advertise `delivery` per node
url4-sdk	`url4 serve` default path `/` ; `Client` default path `/` ; `/v1` alias
url4-sdk	`proxy` mount kind in `url4.toml`
url4-sdk	`Url4Node` → host naming retrofit with a deprecation alias
repo	Doctrine skill synced in this unit (T1, N1, F2, F4, term table)
Kevin	Review Appendix A

Appendix C — where the spec lives

- Parts A and B, v0.5 DRAFT (2026-07-10): `secondbrain/kevin-mcdonough/docs/adrs/URL4-Spec-A.md`, `URL4-Spec-B.md`.
- Parts C–I: "Not yet written" stubs (`…/adrs/site/part-c.html` … `part-i.html`). Their content is the pre-v0.5 monolith, git `8a052dc:kevin-mcdonough/docs/adrs/URL4-Spec.md`, header v0.2. Cited here as "v0.2 §N".
- Public docs: `public-docs/src/pages/learn/Url4Page.vue`. They use "fusion" and "typed DAG"; the spec uses neither.
- Doctrine: `.claude/skills/url4-engine/SKILL.md`, updated with this document.